Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Which of the following statements about recursion are accurate? Select all that apply. 1 / 1 point

- [x] Recursion requires a base case to terminate.

  Nice work

  Correct! Understanding base cases is crucial to avoid infinite recursion.

- [ ] Recursion should execute indefinitely unless a stop condition is manually imposed.
- [x] Recursion can lead to stack overflow if the depth is too great.

  Nice work

  Recursion can lead to stack overflow with too many calls, so manage your recursive depth.

- [ ] Recursion is equivalent to repeating a loop until a condition is met.
- [x] Recursion divides a problem into smaller instances of the same problem.

  Nice work

  Correct! Recursion solves problems by breaking them into smaller sub-problems.

2. When applying the partition algorithm in Quick Sort, if the pivot is always chosen as the first element of the array, what effect might this have on the algorithm's efficiency for certain types of input arrays? 1 / 1 point

- ( ) It results in balanced partitions for all types of input arrays.
- (•) It may cause Quick Sort to degrade to O(n^2) complexity for sorted arrays.
- ( ) It causes Quick Sort to achieve O(log n) efficiency consistently.
- ( ) It improves Quick Sort to best-case efficiency for random arrays.

  Nice work

  Correct! Choosing the first element as the pivot in a sorted array can lead to unbalanced partitions, degrading Quick Sort to O(n^2) complexity.

3. Which of the following characteristics differentiate recursive algorithms from iterative algorithms? Select all that apply. 1 / 1 point

- [ ] Recursive algorithms cannot solve problems iteratively.
- [x] Recursive algorithms use function calls to handle repetitive tasks.

  Nice work

  Correct! Recursive algorithms often rely on function calls to repeat tasks.

- [ ] Recursive algorithms are easier to implement for all problems.
- [x] Iterative algorithms are generally more memory-efficient than recursive ones.

  Nice work

  Correct! Iterative algorithms typically use less memory as they do not require stack space for function calls.

- [ ] Recursive algorithms are always faster than iterative ones.

4. Which of the following statements correctly describe the steps of the Gale-Shapley algorithm for stable matching? 1 / 1 point

- [x] Initiate matching with all participants marked as free.

  Nice work

  Correct! This step ensures that no one is initially matched, setting the stage for the algorithm to run.

- [x] A student rejects a proposal if they have a better offer.

  Nice work

  Correct! The student will hold onto the best offer received, only accepting better ones.

- [ ] The algorithm terminates when no free student is left to propose.
- [x] Each free hospital proposes to its most preferred free student.

  Nice work

  Correct! Hospitals propose to their top-choice students, which is a key action in the algorithm.

- [ ] If a student rejects a hospital, the hospital is eliminated from the process.

5. Which of the following statements accurately describe the merge sort algorithm and its recursive nature? 1 / 1 point

- [x] Merge sort divides the unsorted list into halves, recursively sorts them, and then merges the sorted halves.

  Nice work

  Correct! This describes the divide-and-conquer approach of merge sort.

- [x] Merge sort is a stable sorting algorithm that maintains the relative order of equal elements.

  Nice work

  Correct! Merge sort is indeed stable and maintains the order of equal elements.

- [ ] Merge sort can only sort lists of integers due to its recursive nature.
- [x] Merge sort has a worst-case time complexity of

  $$O(n \log n)$$

  .

  Nice work

  Correct! The divide-and-conquer approach ensures the time complexity is

  $$O(n \log n)$$

  in all cases.

- [ ] The recursive nature of merge sort makes it the most space-efficient sorting algorithm.